OLLAMA

LLM – Complex neural network which have layers & connections

- collection of numbers → [W, B]
  learning (store)

Based on accessibility control over model.

Proprietary model — Open Source Model.

Proprietary models →
- Complete ownership → Company
- Genini → Google
- GPT → OpenAI
- API, control platform
- Raw — [Architecture, ~~Wai~~ weight, Bias]

Open Source Models →
ABC → Model Public → Raw complex
[Archi., Weight, Bias]

Open-source models are AI models whose core details are openly shared with the public.

- These model architecture, weights, and sometimes even the training code are made public. Anyone can download them, study how they work, modify them, fine-tune them for their own needs and run them on their own systems.

- Access: You can download the model files (weights) from platforms like Hugging face and own them for free (though you still pay for the electricity and hardware).
- Customization: Since you have the files, you can "fine-tune" the model on your own private data to make it an expert in a specific field.
- Examples: Meta's Llama 3, Mistral and Deepseek.

Using open source is easy

Models → release → [raw model weight]

Reasons
- ↳ store
- ↳ optimize working memory
- ↳ ~~weight~~ → RAM, VRAM
- ↳ compatible

↓

OLLAMA

Ollama → Ollama is a tool that lets you run large language (LLMs) on your computer. In simple words, Ollama makes it easy to download, run and manage AI models locally using very simple commands.

Benefit

- Privacy and Data control.
- Lower latency & Offline access.
- Cost predictability and potential saving.

① rounded down to the nearest integer → floor.
first → -% , end → %_

- Simple Installation & step
- Pre-Built Model Library.
- Customization of model.
- No vendor lock-in & IP protection.
- Run, Manage Models with simple commands.

Requirements for Using Ollama

① Operating System – Works on macs, Windows, or Linux.

② Hardware – A laptop/PC with at least 8GB RAM (more smoother)

③ Storage Space – Enough free disk space (models can be 3–15 GB each)

④ Internet Connection (first time only) – Needed to download Ollama and models.

⑤ Basic Command Line Knowledge – Knowing how to run simple terminal/command prompt commands.

⑥ (Optional) GPU – Not required, but speeds up performance if available.

Ollama → Download, run, ~~manag~~ manage (Uses)